Choosing Redox Centers for Polymer Flow Batteries

Findings from molecular screening and the next synthesis controls

RTFB Materials Research

October 20, 2026

Molecular design $\rightarrow$ polymer-grafted redox centers

The target is a redox-active polymer for a flow battery

Research objective

Choose pendant redox centers that remain useful after grafting onto the polymer.

Which molecular advantages survive the grafted polymer environment?

Redox potential
Operating window

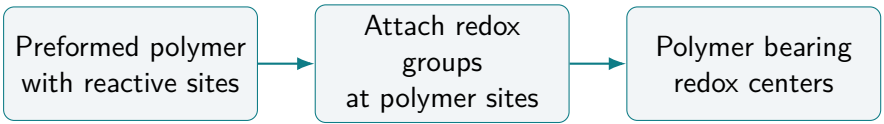
Active loading
Capacity

Charge transfer
Kinetics

Chemical route
Redox retention

Molecular screening supplies design hypotheses; polymer accessibility and cycling must be tested.

Post-functionalize the existing polymer

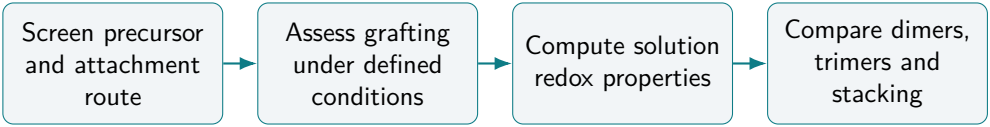


The design question

Which center combines useful redox behavior with a workable route to the polymer?

Potential and molecular capacity guide screening; attachment and accessibility guide downselection.

Workflow: from candidate families to synthesis controls

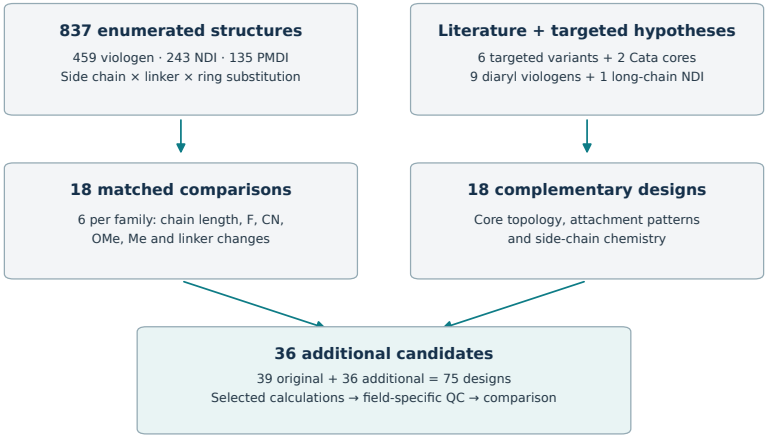


Chemical space
Diimides; benzoquinones (BQ) and anthraquinones (AQ); viologens; positive-side controls.

Selection
Potential, nominal capacity, kinetic evidence and grafting feasibility.

New candidates advance after a reviewed grafting route. Unknown routes stay on hold; method controls remain separate.

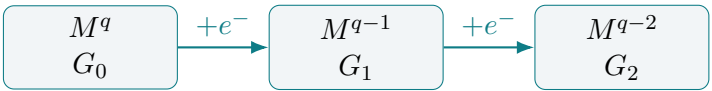
Candidate expansion uses matched molecular changes



Enumeration and selection counts; computational completion is tracked separately.

Current route: attach redox groups to a preformed polymer.

Method: compare free energies of matched solution states



Geometry	r2SCAN-D4 / SVP(D)
Electronic energy	ωB97M-V / TZVPD
Environment	SMD acetonitrile
Thermal correction	Native Hessian, 298 K

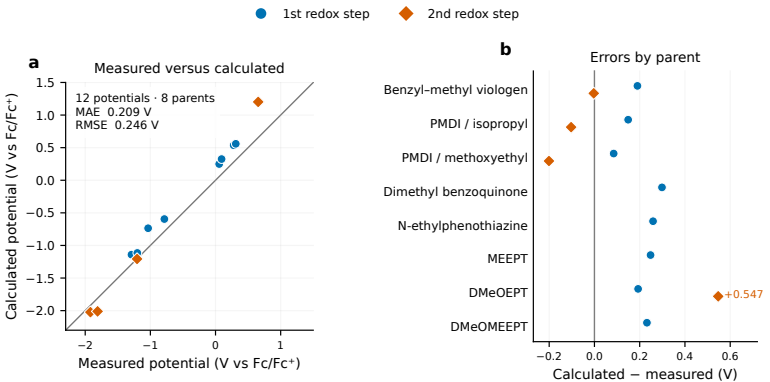
$$G = E_{\text{electronic}} + G_{\text{thermal}}$$

Redox potentials from ΔG, referenced consistently to Fc/Fc^+.

Nominal capacity: $Q = \dfrac{nF}{3.6M}$.

E1/E2 identify successive one-electron steps. Diagram: reductions. Geometry, spin and reference checks are retained.

Validation: actual potentials and model errors



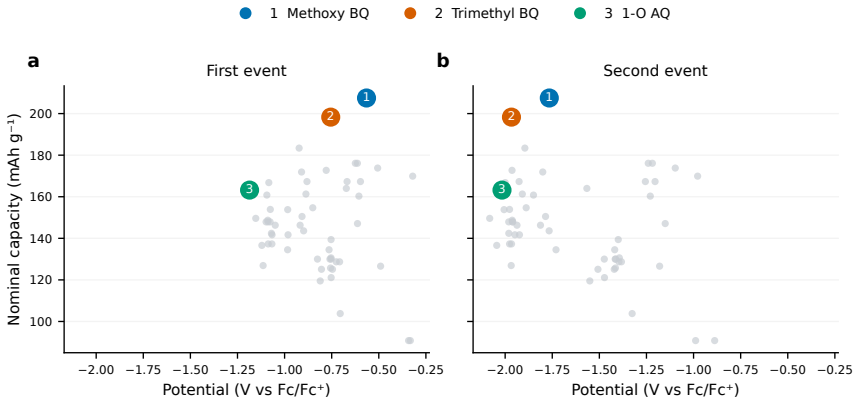
No fitted correction. DMeOEPT second step: measured 0.655 V; calculated 1.202 V.

Selected free-parent benchmark in MeCN; exact grafts, polymer behavior and kinetics are outside this validation.

Observed MAE 0.21 V; RMSE 0.25 V. Largest error: +0.55 V.

Twelve free-parent potentials; exact-graft accuracy and kinetics remain unvalidated.

Quinone models lead nominal capacity

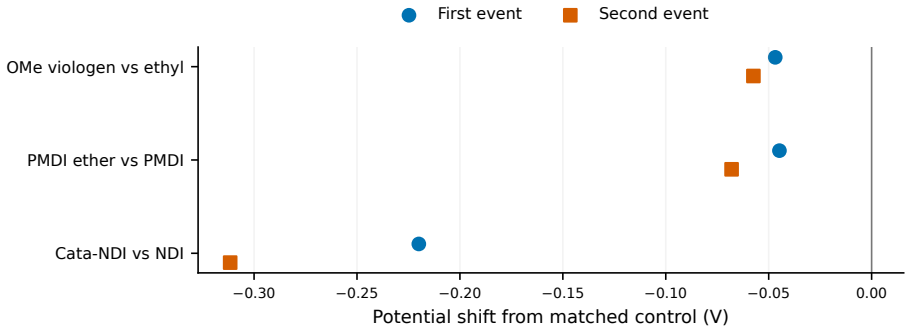


Gray: 56 eligible anolyte models. Numbered candidates are descriptor leaders, not synthesis recommendations.

The two BQ models reach 207.5 and 198.3 mAh g^{-1}.

Base required; BQ retained. DBU: -45.1/-46.7 kJ mol^{-1} (MeCN model); selectivity and yield untested.

Core topology changes NDI reduction potentials

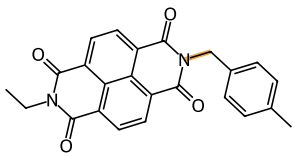


NDI core topology shifts E_1 by -0.220 V and E_2 by -0.312 V.

Matched model differences in MeCN; voltage shifts alone do not establish battery performance.

Cata-NDI shifts potential without a nominal capacity penalty

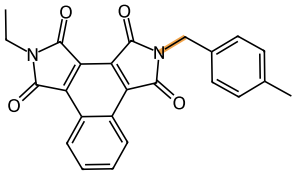
Conventional NDI / ethyl



E1: -0.764 V | E2: -1.418 V

Nominal capacity: 134.5 mAh g^{-1}

Cata-NDI / ethyl



E1: -0.984 V | E2: -1.730 V

Nominal capacity: 134.5 mAh g^{-1}

ΔE_1 = -0.220 V · ΔE_2 = -0.312 V

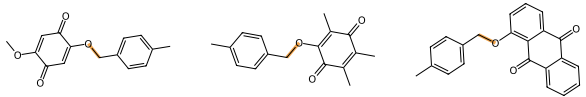
Same molecular formula. Potentials vs Fc/Fc^+ in MeCN. Local graft models; polymer effects are separate.

A core-isomer comparison worth discussing alongside the capacity leaders.

Local graft models; precursor availability, attachment and polymer behavior remain untested.

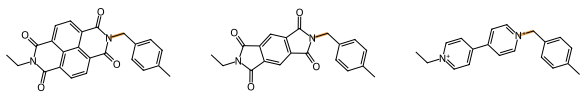
Capacity leaders and representative graft-site controls

a Descriptor leaders · computed capped models



1 Methoxy BQ | 2 Trimethyl BQ | 3 1-O AQ

b Representative graft-site controls · computed capped models



C1 NDI graft-site model | C2 PMDI graft-site model | C3 Viologen graft-site model

Orange bond: attachment to the redox center. These are local models of grafted sites.

Base required; BQ candidates retained for evaluation.

NDI: naphthalenediimide; PMDI: pyromellitic diimide. Polymer effects unvalidated.

Downselection couples redox behavior with grafting feasibility

Attach	Load	Operate
Precursor availability	Degree of grafting	Swelling and electrolyte
Attachment selectivity	Accessible active sites	Charge transport
Redox-center retention	Counterion mass	Cycling stability

High nominal molecular capacity is a screening descriptor; usable polymer capacity depends on accessible grafted sites.

Local benzyl-capped calculations do not include the full polymer environment.

Three findings guide the next experiments

Capacity

Substituted BQ models lead the current nominal-capacity screen.

Voltage tuning

NDI core topology provides a clear matched computational contrast.

Synthesis direction

Base required: retain BQ candidates and test selective attachment.

Next decision: combine molecular redox properties with attachment feasibility, graft loading and accessible redox sites. Solubility and cycling performance remain unmeasured for these exact targets.

Backup

Methods, validation and detailed comparisons

Grafting: define the reaction before ranking candidates

Local attachment models in acetonitrile, 298.15 K

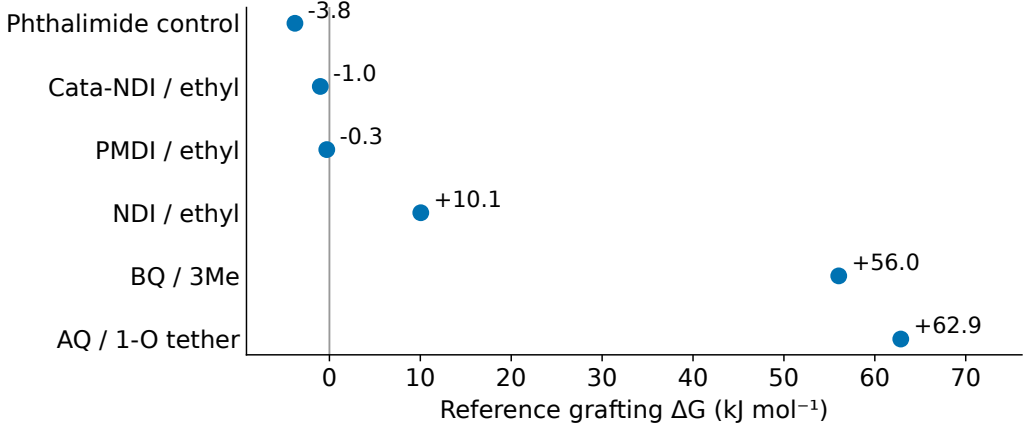
N–H / O–H: $\mathrm{site{-}Cl + Nu{-}H \rightarrow site{-}Nu + HCl}$

N quaternization: $\mathrm{site{-}Cl + Nu \rightarrow site{-}Nu^{+} + Cl^{-}}$

Compare within each reaction class. An explicit base contribution is still pending.

These are capped-site thermodynamic models. Activation barriers, yields, site accessibility and polymer interactions are separate questions.

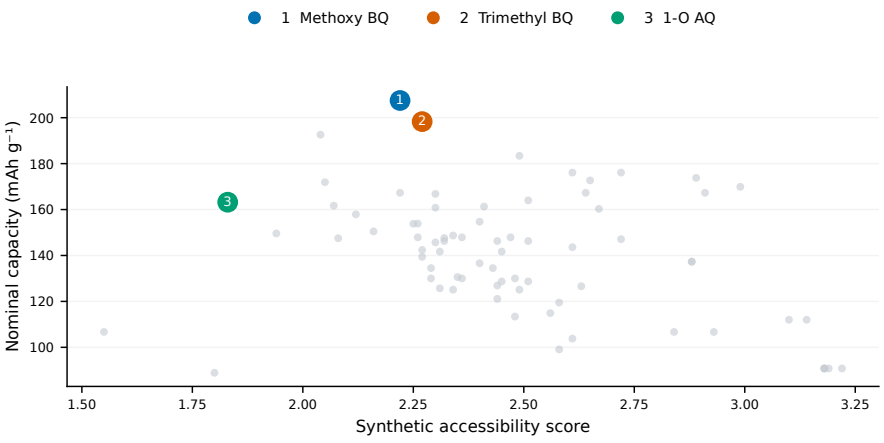
Grafting thermodynamics: the first computed results



N–H/O–H reference: $\mathrm{site{-}Cl + Nu{-}H \rightarrow site{-}Nu + HCl}$. Base capture is uncomputed; values do not predict grafting rates or yields.

10/75 reactions qualified at 2026-10-19 20:42 Chicago; continued calculations update the visual report.

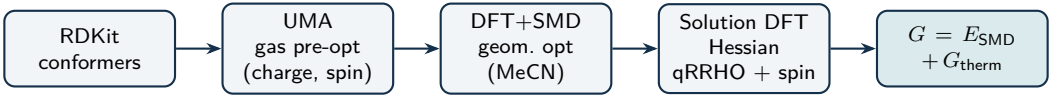
SA and nominal capacity: a supplementary comparison



Gray: other designs. Lower SA is a heuristic, not evidence of a polymer synthesis route.

SA is a molecular heuristic; synthesis and polymerization compatibility require independent review.

The pipeline (per redox state)



Level of theory

- opt: r^{2}SCAN-D4/def2-SVP(D) + SMD
- energy: ωB97M-V/def2-TZVPD (all charges)
- referenced to level-matched Fc/Fc$^+$

Checks before export

- Full grid response and matching CPHF grids; finite-gradient checks for RKS, UKS and Fc
- Current geometry/spin; matched native Fc thermal definition; incomplete properties withheld

How each axis is computed

axis	method	validation / trust
redox potential	DFT+SMD ΔG; ωB97M-V/def2-TZVPD // r^{2}SCAN/def2-SVP(D); live level-matched Fc	Matched native Fc; 7 parent events, observed RMSE 0.171 V
thermal G_{therm}	Solution DFT Hessian, projected internal-mode qRRHO + explicit spin (298 K)	Independent derivative and minimum checks
stability (disprop.)	$\Delta G_{\text{disp}} = F(E_{\text{hi}} - E_{\text{lo}})$ from state G's	Same thermal definition across all three states
stability (dimer.)	pimer cluster DFT+SMD ΔG	**flagged** (+2 desolvation)
reorg. energy λ	Gas Nelsen diagnostic; native SMD frozen-slow directional relaxation	Minimum, conformer and experimental controls required
integrity gate	bond-graph change + solvated EA	necessary, not sufficient for reversibility
capacity	$nF/(3.6\,\text{MW})$ in mAh/g (active species)	Stoichiometry; accessible n requires validated potentials
solubility	$\Delta G_{\text{solv}} = E_{\text{SMD}} - E_{\text{gas}}$ (neutral)	proxy (relative)
synth. access.	RDKit SA_Score	heuristic proxy

Geometries optimized in SMD(MeCN); conformers via RDKit + UMA pre-opt.

Candidate calculations: current verified snapshot

39 original proposals + 36 additional candidates.

- 62 structures with completed, checked native potential ladders.
- 115 original-candidate DFT geometry states completed.
- Independent arithmetic: 628 checks, zero discrepancies.
- Pending or flagged fields remain blank.

Data snapshot: 2026-10-19 18:29:30 (America/Chicago).

What the numbers establish

Current geometry, spin evidence, active SMD energies and derivative-checked solution thermochemistry.

Predicted molecular properties; experimental accuracy for the exact grafts remains unvalidated.

Native frequency gate: $\nu_{\min} \geq -5$ cm^{-1}. Small negative modes within this tolerance remain flagged in the audit.

Candidate expansion: matched computational hypotheses

Design	Nominal capacity mAh/g	UMA states completed	Native states admitted
PMDI / cyclopropyl	148.7	3/3	3/3
PMDI / 2fluoroethyl	146.3	3/3	3/3
NDI / cyclopropyl	130.6	3/3	3/3
NDI / 2fluoroethyl	128.7	3/3	3/3
PTZ 3,7-dimethyl	161.7	3/3	3/3
PTZ 3,7-difluoro	157.9	3/3	3/3
Cata-NDI ethyl	134.5	3/3	3/3
Cata-ADI ethyl	119.5	3/3	3/3

Nominal capacity assumes the declared two-electron ladder. Usable electrons require both potential-window and integrity gates.

UMA is a gas-phase preoptimizer. New solution potentials, disproportionation and solution kinetics remain pending until their separate checks pass.

Exact-graft measurements and synthesis routes are unverified. Parent results are not assigned to these structures.

Audited snapshot: 2026-10-19 18:29:30.

New computational leads for experimental follow-up

Exact graft	E_1	E_2	Nominal capacity
	V vs Fc/Fc^+		mAh/g
3,7-difluoro PTZ	0.689	1.696	157.9
Ether-linked PMDI	-1.121	-2.045	136.6
Ring-OMe viologen	-0.668	-1.256	167.3
Cyclopropyl PMDI	-1.089	-1.962	148.7

Fluoro PTZ is a positive-side voltage hypothesis; its second wave and usable capacity remain provisional under the parent model limitation.

OMe viologen supplies a matched donor-ring contrast; compact PMDI retains useful nominal capacity. Small shifts are unresolved improvements.

This is a follow-up set, not a validated performance ranking. Exact-graft solubility, cycling durability, synthesis and solution kinetics remain unvalidated.

Corrected native snapshot: 2026-10-19 18:29:30.

Core-topology contrast: cata-NDI and conventional NDI

Exact graft	E_1	E_2	Nominal capacity
	V vs Fc/Fc^+		mAh/g
Conventional NDI / Et	-0.764	-1.418	134.5
Cata-NDI / Et	-0.984	-1.730	134.5

Same N-ethyl cap and Merrifield N-graft anchor. This isolates the core-topology isomer contrast.

Candidate minus control: $\delta E_1 = -0.220$ V; $\delta E_2 = -0.312$ V. The shared Fc reference cancels.

A more negative anolyte voltage is a lead for follow-up. Overall battery performance, exact synthesis, solubility and solution kinetics remain unvalidated.

Current calculated graft properties; CH_2Cl_2 source-parent measurements are separate.

Audited snapshot: 2026-10-19 18:29:30.

Fused-ring extension: cata-NDI to cata-ADI

Exact graft	E_1	E_2	Nominal capacity
	V vs Fc/Fc^+		mAh/g
Cata-NDI / Et	-0.984	-1.730	134.5
Cata-ADI / Et	-0.811	-1.551	119.5

Same N-ethyl cap and Merrifield anchor. The added fused ring changes both core size and mass.

The voltage/capacity tradeoff is calculated at one solution protocol with field-specific geometry, derivative and spin checks.

The cata-ADI neutral required a reviewed finer CDS derivative step. Its original failed Hessian is preserved; acceptance limits and electronic methods are unchanged.

Exact-graft synthesis, solubility, durability and solution rates remain unvalidated.

Audited snapshot: 2026-10-19 18:29:30.

Systematic family screening: comparison panels

Family	Comparison	UMA states	Native states
Viologen	Propyl cap	3/3	3/3
Viologen	Ring F	3/3	3/3
Viologen	Ring CN	3/3	3/3
Viologen	Ether linker	3/3	3/3
PMDI	Propyl cap	3/3	3/3
PMDI	Ring F	3/3	3/3
PMDI	Ring CN	3/3	3/3
PMDI	Ether linker	3/3	3/3
NDI	Propyl cap	3/3	3/3
NDI	Ring F	3/3	3/3
NDI	Ring CN	3/3	3/3
NDI	Ether linker	3/3	3/3

Matched ethyl controls; all ring variants use the same selected site.

UMA is gas-phase screening. Solution potentials and usable electrons require DFT, thermal, spin and integrity QC.

Exact synthesis routes and experimental graft performance remain unverified.

Audited snapshot: 2026-10-19 18:29:30.

Systematic family screening: comparison panels

Family	Comparison	UMA states	Native states
Viologen	Ring OMe	3/3	3/3
Viologen	Ring Me	3/3	3/3
PMDI	Ring OMe	3/3	3/3
PMDI	Ring Me	3/3	3/3
NDI	Ring OMe	3/3	3/3
NDI	Ring Me	3/3	3/3

Matched ethyl controls; all ring variants use the same selected site.

UMA is gas-phase screening. Solution potentials and usable electrons require DFT, thermal, spin and integrity QC.

Exact synthesis routes and experimental graft performance remain unverified.

Audited snapshot: 2026-10-19 18:29:30.

Matched family comparisons: admitted potential shifts

Family	Contrast	δE_1	δE_2	$\delta(\Delta E)$
		Candidate minus declared control (V)		
Viologen	Propyl cap	0.008	-0.020	0.028
Viologen	Ring F	0.115	0.101	0.014
Viologen	Ring CN	0.301	0.220	0.081
Viologen	Ether linker	0.016	-0.031	0.047
PMDI	Propyl cap	-0.022	0.015	-0.037
PMDI	Ring F	0.158	0.164	-0.006
PMDI	Ring CN	0.177	0.210	-0.033
PMDI	Ether linker	-0.045	-0.068	0.023
NDI	Propyl cap	0.011	0.005	0.006
NDI	Ring F	0.056	0.036	0.020

Each shift requires an admitted candidate and control event under matched native thermochemistry. Pending events stay blank.

Positive potential shifts are more positive reductions; the shared Fc reference cancels. A shift alone does not establish improved battery performance.

Audited snapshot: 2026-10-19 18:29:30.

Matched family comparisons: admitted potential shifts

Family	Contrast	δE_1	δE_2	$\delta(\Delta E)$
		Candidate minus declared control (V)		
NDI	Ring CN	0.274	0.238	0.035
NDI	Ether linker	0.011	-0.054	0.065
Viologen	Ring OMe	-0.047	-0.057	0.011
Viologen	Ring Me	-0.003	-0.042	0.039
PMDI	Ring OMe	0.094	0.053	0.041
PMDI	Ring Me	-0.012	0.017	-0.029
NDI	Ring OMe	-0.040	-0.089	0.049
NDI	Ring Me	-0.061	-0.055	-0.006

Each shift requires an admitted candidate and control event under matched native thermochemistry. Pending events stay blank.

Positive potential shifts are more positive reductions; the shared Fc reference cancels. A shift alone does not establish improved battery performance.

Audited snapshot: 2026-10-19 18:29:30.

New candidate potentials with the matched native Fc reference

Candidate	E_1	E_2	ΔE	Nominal	Accessible
	V vs. Fc/Fc^+			mAh/g	
PTZ 3,7-dimethyl	0.480	1.460	0.980	161.7	–
PTZ 3,7-difluoro	0.689	1.696	1.008	157.9	–
PMDI ether linker	-1.121	-2.045	0.924	136.6	136.6
PMDI MEE cap	-1.114	-1.968	0.854	126.9	126.9
PMDI propyl cap	-1.098	-1.962	0.863	147.9	147.9
PMDI cyclopropyl	-1.089	-1.962	0.873	148.7	148.7
PMDI ring Me	-1.088	-1.959	0.871	147.9	147.9
PMDI (R)-pentan-2-yl	-1.087	-1.966	0.879	137.3	137.3

Active-species mass; no counterion/backbone correction. Accessible capacity uses the model window and integrity path; scoped model limits leave it blank. Parent residuals do not validate exact grafts.

Solution kinetics and synthetic feasibility require separate checks.

Audited snapshot: 2026-10-19 18:29:30.

New candidate potentials with the matched native Fc reference

Candidate	E_1	E_2	ΔE	Nominal	Accessible
	V vs. Fc/Fc^+			mAh/g	
PMDI iPr cap	-1.077	-1.981	0.904	147.9	147.9
PMDI Et cap	-1.076	-1.977	0.901	153.9	153.9
PMDI n-butyl cap	-1.070	-1.981	0.911	142.4	142.4
PMDI (S)-pentan-2-yl	-1.067	-1.977	0.909	137.3	137.3
PMDI ME cap	-1.067	-1.948	0.882	141.7	141.7
PMDI 2-fluoroethyl	-1.050	-1.937	0.887	146.3	146.3
Cata-NDI ethyl	-0.984	-1.730	0.746	134.5	134.5
PMDI ring OMe	-0.982	-1.924	0.942	141.7	141.7

Active-species mass; no counterion/backbone correction. Accessible capacity uses the model window and integrity path; scoped model limits leave it blank. Parent residuals do not validate exact grafts.

Solution kinetics and synthetic feasibility require separate checks.

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New candidate potentials with the matched native Fc reference

Candidate	E_1	E_2	ΔE	Nominal	Accessible
	V vs. Fc/Fc^+			mAh/g	
PMDI ring F	-0.918	-1.813	0.895	146.3	146.3
PMDI ring CN	-0.899	-1.767	0.868	143.6	143.6
NDI ring Me	-0.825	-1.474	0.648	130.0	130.0
Cata-ADI ethyl	-0.811	-1.551	0.740	119.5	119.5
NDI ring OMe	-0.804	-1.507	0.703	125.1	125.1
NDI Et cap	-0.764	-1.418	0.654	134.5	134.5
NDI iPr cap	-0.759	-1.416	0.657	130.0	130.0
NDI n-butyl cap	-0.756	-1.415	0.659	125.7	125.7

Active-species mass; no counterion/backbone correction. Accessible capacity uses the model window and integrity path; scoped model limits leave it blank. Parent residuals do not validate exact grafts.

Solution kinetics and synthetic feasibility require separate checks.

Audited snapshot: 2026-10-19 18:29:30.

New candidate potentials with the matched native Fc reference

Candidate	E_1	E_2	ΔE	Nominal	Accessible
	V vs. Fc/Fc^+			mAh/g	
NDI cyclopropyl	-0.754	-1.394	0.640	130.6	130.6
NDI propyl cap	-0.753	-1.413	0.660	130.0	130.0
NDI Me cap	-0.753	-1.400	0.646	139.4	139.4
NDI ether linker	-0.753	-1.473	0.720	121.1	121.1
NDI ME cap	-0.746	-1.421	0.675	125.1	125.1
NDI 2-fluoroethyl	-0.726	-1.396	0.670	128.7	128.7
NDI ring F	-0.708	-1.383	0.675	128.7	128.7
NDI MEEE cap	-0.705	-1.326	0.621	103.8	103.8

Active-species mass; no counterion/backbone correction. Accessible capacity uses the model window and integrity path; scoped model limits leave it blank. Parent residuals do not validate exact grafts.

Solution kinetics and synthetic feasibility require separate checks.

Audited snapshot: 2026-10-19 18:29:30.

New candidate potentials with the matched native Fc reference

Candidate	E_1	E_2	ΔE	Nominal	Accessible
	V vs. Fc/Fc^+			mAh/g	
NDI ring CN	-0.490	-1.180	0.690	126.6	126.6
Phthalimide (1e control)	-1.642	–	–	106.7	106.7
Naphthalimide (1e control)	-1.522	–	–	88.9	88.9
3,7-diOMe PTZ (2e proposal)	0.345	1.279	0.934	147.5	–
Viologen ring OMe	-0.668	-1.256	0.587	167.3	167.3
Viologen ring Me	-0.624	-1.241	0.616	176.1	176.1
Viologen propyl cap	-0.614	-1.219	0.605	176.1	176.1
Viologen MEE cap	-0.613	-1.150	0.537	147.1	147.1

Active-species mass; no counterion/backbone correction. Accessible capacity uses the model window and integrity path; scoped model limits leave it blank. Parent residuals do not validate exact grafts.

Solution kinetics and synthetic feasibility require separate checks.

Audited snapshot: 2026-10-19 18:29:30.

New candidate potentials with the matched native Fc reference

Candidate	E_1	E_2	ΔE	Nominal	Accessible
	V vs. Fc/Fc^+			mAh/g	
Viologen ether linker	-0.605	-1.230	0.624	160.3	160.3
Viologen ME cap	-0.597	-1.204	0.607	167.3	167.3
Viologen ring F	-0.506	-1.097	0.591	173.8	173.8
AV-3c ester at aryl C3/5	-0.342	-0.888	0.546	90.8	90.8
AV-3e ester at aryl C2	-0.333	-0.989	0.656	90.8	90.8
Viologen ring CN	-0.320	-0.978	0.658	169.9	169.9
AQ 1-O tether	-1.186	-2.017	0.831	163.2	163.2
AQ 2-O tether / 6-OMe	-1.153	-2.082	0.929	149.6	149.6

Active-species mass; no counterion/backbone correction. Accessible capacity uses the model window and integrity path; scoped model limits leave it blank. Parent residuals do not validate exact grafts.

Solution kinetics and synthetic feasibility require separate checks.

Audited snapshot: 2026-10-19 18:29:30.

New candidate potentials with the matched native Fc reference

Candidate	E_1	E_2	ΔE	Nominal	Accessible
	V vs. Fc/Fc^+			mAh/g	
D3TaLES 80NOPT	-1.094	-1.850	0.755	160.8	160.8
D3TaLES 80NECN	-1.084	-2.000	0.916	166.8	166.8
D3TaLES 80EEOZ	-0.984	-2.006	1.022	153.8	153.8
D3TaLES 80XWVS	-0.923	-1.895	0.973	183.4	183.4
D3TaLES 05TXCN	-0.910	-1.800	0.890	171.9	171.9
D3TaLES 05MNMT	-0.907	-1.786	0.878	150.5	150.5
D3TaLES 80GKVH	-0.887	-1.909	1.022	161.3	161.3
D3TaLES 80GXWG	-0.881	-1.926	1.045	167.3	167.3

Active-species mass; no counterion/backbone correction. Accessible capacity uses the model window and integrity path; scoped model limits leave it blank. Parent residuals do not validate exact grafts.

Solution kinetics and synthetic feasibility require separate checks.

Audited snapshot: 2026-10-19 18:29:30.

New candidate potentials with the matched native Fc reference

Candidate	E_1	E_2	ΔE	Nominal	Accessible
	V vs. Fc/Fc^+			mAh/g	
D3TaLES 80LEVD	-0.850	-1.888	1.038	154.7	154.7
D3TaLES 80TBWR	-0.779	-1.964	1.184	172.7	172.7
BQ O tether / 3Me	-0.755	-1.967	1.211	198.3	198.3
D3TaLES 80MDPM	-0.672	-1.566	0.894	164.0	164.0
BQ O tether / OMe	-0.565	-1.766	1.201	207.5	207.5
D3TaLES 05CZHI	-0.406	-1.313	0.906	147.6	73.8

Active-species mass; no counterion/backbone correction. Accessible capacity uses the model window and integrity path; scoped model limits leave it blank. Parent residuals do not validate exact grafts.

Solution kinetics and synthetic feasibility require separate checks.

Audited snapshot: 2026-10-19 18:29:30.

Reference and thermal validation completed

Fc geometries repaired on the solution DFT surface.

State	Earlier	Repaired
Fc	-35.1	22.5
Fc^+	-21.7	18.9

Lowest DFT vibrational frequency (cm^{-1}). Both current references are stationary minima.

- Full grid response and matching CPHF grids verified by independent finite gradients.
- Matched solution thermochemistry at 298.15 K, including the explicit spin partition.
- Five Fe occupation guesses reproduce the chosen geometry-level electronic solution.

Sources: native thermal certificate, matched Fc reference and occupation-control records (2026-10-11).

Promising computational leads: a diversified test shortlist

Candidate	E_1	E_2	Capacity
	V vs Fc/Fc^+		mAh/g
AQ 1-O tether	-1.186	-2.017	163.2
PMDI Et cap	-1.076	-1.977	153.9
NDI Me cap	-0.753	-1.400	139.4
Viologen ME cap	-0.597	-1.204	167.3
BQ O tether / 3Me	-0.755	-1.967	198.3

Compact AQ/diimide/viologen branches plus a conditional high-capacity BQ branch. Capacity is active-species bookkeeping.

Exact-graft experimental performance remains unvalidated; synthesis, solution kinetics and durability are required next gates.

The trends are plausible; the apparent gains need qualification

- Longer caps add mass and reduce gravimetric capacity; transport benefits have not been demonstrated.
- NDI and PMDI give distinct core/voltage tradeoffs; millivolt alkyl-cap differences are not resolved improvements.
- $\Delta G_{disp} = F\Delta E$: wave spacing and disproportionation reuse the same free-energy information.
- Two potentials do not establish two usable electrons: D3TaLES 05CZHI passes only one step under bond-integrity QC.

Historical points use superseded reference/thermal definitions. Overall superiority over earlier candidates is unestablished.

PTZ first couple: solution relaxation diagnostic

Difluorophenothiazine graft: $\mathrm{ox1}(+1) + e^- \rightleftharpoons \mathrm{neu}(0)$ in MeCN.

Electronic contribution (eV)	Reduction	Oxidation
Equilibrium geometry relaxation	0.202	0.223
Frozen slow-solvent increment	0.914	0.915
Directional solution relaxation	1.116	1.137

Local pairing checked; kinetic qualification remains incomplete

Reciprocal full-response optimizations return both selected local geometries (heavy-atom RMSDs below 0.0018 Å). These are qualified composite electronic descriptors at geometry-level minima; active-SP stationarity, thermal activation, global conformer populations and polymer constraints remain unproven. Primary kinetic rates remain pending.

Native frozen-slow SMD response; no addition of a separate gas or outer-sphere term.

Updated 19 October 2026: six archived densities rechecked; 16 arithmetic checks and 10 reciprocal-geometry checks pass. Earlier 99-check audit preserved. Exact-graft experimental accuracy remains unvalidated.

Historical parent benchmark: superseded thermochemistry

Archived 18 October snapshot: superseded thermal partition and Fc reference.

Parent molecule	E_1 error	E_2 error	Wave-spacing error
	calculated minus experimental (V)		
Benzyl–methyl viologen	$+0.197$	$+0.038$	$+0.159$
PMDI / isopropyl	$+0.166$	-0.105	$+0.271$
PMDI / methoxyethyl	$+0.112$	-0.216	$+0.328$
Dimethyl benzoquinone	$+0.3085$	—	—

Historical potential RMSE: **0.183 V**; MAE: 0.163 V.

Wave spacing still needs model validation

All three spacings are overestimated. Fc cancels. Electronic gaps exceed experiment; PMDI thermal terms widen them further. No correction was fitted.

Matched 25–100 cm^{-1} cutoff scan: potential span $\leq$0.036 V; separation bias persists.

These numbers are retained for provenance, not current calibration. The corrected, spin-qualified comparison is on the next slide. Supporting-electrolyte effects are not explicitly modeled; the PMDI source leaves the salt assignment ambiguous.

Measured and calculated parent potentials

Parent	First redox step		Second redox step	
	Measured	Calculated	Measured	Calculated
Benzyl–methyl viologen	-0.785	-0.595	-1.204	-1.207
PMDI / isopropyl	-1.290	-1.141	-1.920	-2.023
PMDI / methoxyethyl	-1.200	-1.115	-1.810	-2.011
Dimethyl benzoquinone	-1.0345	-0.736	—	—
N-ethylphenothiazine	0.278	0.537	—	—
MEEPT	0.311	0.559	—	—
DMeOEPT	0.056	0.249	0.655	1.202
DMeOMEEPT	0.093	0.325	—	—

All values V vs Fc/Fc^+ in MeCN. First/second refer to successive reductions for imides, quinone and viologen; successive oxidations for phenothiazines.

Dashes: no admitted measurement for that step; excluded waves are not plotted. MAE 0.209 V; RMSE 0.246 V. No correction fitted.

Twelve potentials, eight exact free parents, four original studies. Source snapshot: 19 October 2026.

Phenothiazine validation: second-wave model limitation

Parent	Exp. E_1	Calc. E_1	Exp. E_2	Calc. E_2
EPT	0.278	0.537	$0.956^{\dagger}$	—
MEEPT	0.311	0.559	$0.937^{\dagger}$	—
DMeOEPT	0.056	0.249	0.655	1.202
DMeOMEEPT	0.093	0.325	$0.666^{\dagger}$	—
B(MEEO)EPT	$0.064^{\ddagger}$	—	$0.654^{\ddagger}$	—

- All values V vs Fc. Measurements: 1 mM, 0.5 M TEATFSI/MeCN, $26\,^{\circ}$C, 10 mV/s; authors' ferrocene calibration.
- DMeOEPT charging gap: 0.953 V calculated vs 0.599 V measured. Electronic / thermal terms: 0.914 / 0.039 eV; Fc cancels from this gap.
- No lower state found in four zero-net-spin orbital-seed probes. No canonical replacement or fitted correction; model diagnosis remains open.

† Second wave excluded: chemical reversibility not established. ‡ Table/discussion conflict; tier B. Attanayake et al., Chem. Mater. 2019, DOI: 10.1021/acs.chemmater.8b04770, Table 1. Parent evidence does not validate grafts.

Phenothiazine model diagnosis: geometry sensitivity

DMeOEPT state	Max. force Ha/Bohr	FD error	Local δE estimate eV
Neutral	0.01555	1.96×10^{-6}	-0.080
Radical cation	0.01490	1.74×10^{-6}	-0.090
Dication	0.01493	0.84×10^{-6}	-0.109

- Three independent reserved H100 calculations reproduce selected energies; the largest gradient component passes a cold finite-energy check.
- The declared composite geometries are not stationary at the active level. A local lower-level Hessian estimate shifts the charging gap by only -0.0084 eV.
- This is neither an optimized free-energy result nor an error bound. Functional, environment and geometry contributions remain unresolved.

wB97M-V/TZVPD/SMD forces at r2SCAN-D4 geometries; local trust radius 0.1 Å. No canonical energy, geometry or thermal term was replaced.

Observed model limits qualify candidate selection

The second-oxidation prediction is provisional

A checked free parent has a $+0.547$ V second-wave residual. Phenothiazine $+2/+1$ catholyte paths therefore require further model validation.

- Retain calculated potentials and nominal electron/mass capacities.
- Withhold qualified accessible capacity for three PTZ candidates; retain the original window/path estimate as a separately labelled diagnostic.
- Exclude affected voltage, capacity and stability objectives from primary Pareto selection. First-wave-only paths and other molecular cores keep their QC.
- Keep the poor-agreement point in the experimental benchmark. No offset fit or parent measurement is assigned to an exact graft.

Evidence-bound model scope is independent of solver convergence, derivative checks and arithmetic agreement. Exact-graft performance remains unvalidated.

Phenothiazine diagnosis: active-level relaxation

DMeOEPT state	Electronic relaxation eV	Final max. gradient Ha/Bohr
Neutral	-0.1004	2.10e-06
Radical cation	-0.0925	3.44e-06
Dication	-0.1116	2.60e-06

- Actual wB97M-V/TZVPD/SMD relaxation changes the electronic charging gap from 0.913645 to 0.886511 eV: $\delta = -0.027134$ eV.
- Optimizer, cold-SCF, stationary-gradient, graph and finite-energy checks pass. Independent review: 37 checks, zero failures.
- This local electronic geometry effect is smaller than the original 0.354-eV free-energy-gap discrepancy. Other model contributions remain unresolved.

Private sensitivity: thermals were not recomputed and Hessian minima were not proved. No canonical replacement or fitted correction; do not silently reuse old thermals.

Uncertainty evidence: computational spread and priors

The lambda scale does not validate solution kinetics

Current scale: 0.209 eV from 330 selected computational-model residuals. The median residual is -0.100 eV; no offset is fitted or applied.

- The source comparison has no convergence, binding or conformer-QC columns. A value-range filter does not certify those properties.
- At least two comparisons are required to estimate a spread. Sparse or zero-spread comparisons cannot tighten the configured 0.10-eV prior floor.
- Evidence type, counts, spread and location shift are exported explicitly. Three regression cases pass; 544 software tests passed at that checkpoint.
- Current numeric property/scorecard/Pareto tables are unchanged. Empirical solution-rate and exact-graft accuracy remain unvalidated.

Computational-reference discrepancy is separate from empirical predictive error. Existing field-specific gas, spin and physical-QC exclusions remain active.

Phenothiazine diagnosis: electronic-method sensitivity

Electronic approximation	Charging gap eV	Change from WB eV
wB97M-V	0.886511	0.000000
PBE0-D4	0.954183	+0.067671
B3LYP-D4	0.908061	+0.021550

- Nine predeclared calculations at the same three WB-relaxed geometries; def2-TZVPD, RI-J, SMD/MeCN and grid 3 throughout.
- All points pass SCF/spin checks. Three WB controls reproduce prior energies; independent review: 71 checks, zero failures. Eight reserved H100 workers ended.
- These alternatives raise the electronic gap. They do not explain the original positive second-wave residual; the PTZ selection hold remains.

Functionals and dispersion differ together. Fixed-geometry electronic sensitivity is not a new free energy or empirical error bar. No method selected, fitted correction, old-thermal reuse or canonical replacement.

Diaryl-viologen expansion: qualified gas-screen inputs

- Nine source-directed O-grafts retain both N–aryl bonds; carboxyl positions and two phenolic attachment sites extend the alkyl-cap comparisons.
- All 27 declared charge states and 270 conformer/state input checks pass. Three 70-atom T4 pilot states pass: peak device memory 1473 MiB.
- All 27 gas UMA states passed independent terminal review and were installed byte-identically for the separate MeCN campaign on 18 suitable reserved GPUs.
- Frozen acid/phenol stoichiometry is a hypothesis. MeCN equilibrium speciation, exact monoattachment synthesis and solution potentials remain unestablished.

Explicit hydrogen attachment review

798 stored geometries screened. Three quinone dianions change H attachments in both stages; their second reductions already fail bond-integrity gates. Metal and ion-pair graphs remain separately scoped. All 558 software tests pass.

Gas warm starts are not MeCN potentials or validated battery performance. Parent aqueous/pH14/AgAgCl measurements are separate. No DFT/Hessian qualification for 16-GB cards.

Registered diaryl-viologen candidates: solution stage

Exact graft	Gas UMA	Checked native states	MeCN potentials
AV-3a ester at aryl C3	3/3	0/3	pending
AV-3b ester at aryl C4	3/3	0/3	pending
AV-3c ester at aryl C3/5	3/3	0/3	pending
AV-3d ester at aryl C3	3/3	1/3	pending
AV-3d ester at aryl C4	3/3	2/3	pending
AV-3e ester at aryl C2	3/3	2/3	pending
AV-3e ester at aryl C3	3/3	0/3	pending
AV-3f ether at aryl C2	3/3	0/3	pending
AV-3f ether at aryl C5	3/3	0/3	pending

All nine graphs have the declared $+2/+1/0$ ladder. Potentials and usable capacities need complete geometry, spin, thermal and integrity admission. Parent measurements stay separate.

Five selected states pass 303 independent checks (19 Oct, 12:07 CDT). Memory-deferred states await H100s; completed stages and physical thresholds are retained.

Hexyl NDI: contact diagnosis and distinct basins

Persistent CDS force jump

H/H solvent-sphere contact	3.2000001 Å
Full force jump	2.1983×10^{-4} Ha/Bohr
Predicted FD residual	0.219830 Ha/Bohr^2
Preserved quantum residual	0.219800 Ha/Bohr^2
Relative discrepancy	0.01346%

38 independent diagnostic checks pass. Both previous geometries remain unqualified.

Two different starting basins

Seed	Heavy RMSD	Torsions changed
08	4.307 Å	6
09	5.029 Å	7

149 independent preparation checks pass. Native-level optimization, active energies and thermochemistry are pending.

Admission stays pending: compare energy ordering and conformer scope; a higher passing basin cannot replace an unresolved lower state by itself.

Evidence: hexyl_conformer_search_2026-10-19/contact_independent_review.json, independent_review.json and native_seed_plan.json.

Source-directed expansion: diaryl viologens with O-grafts

Seven named parent structures curated from synthesis and characterization data.

Prepared branch	Exact mono-O-grafts
Mono-carboxyphenyl positional variants	2
Dicarboxyphenyl positional variants	5
2,5-Dihydroxyphenyl: C2/C5 O-sites	2

Both N–aryl bonds are retained. Attachment is explored through carboxyl benzyl esters or phenolic benzyl ethers; PV is the parent with no supported O-handle.

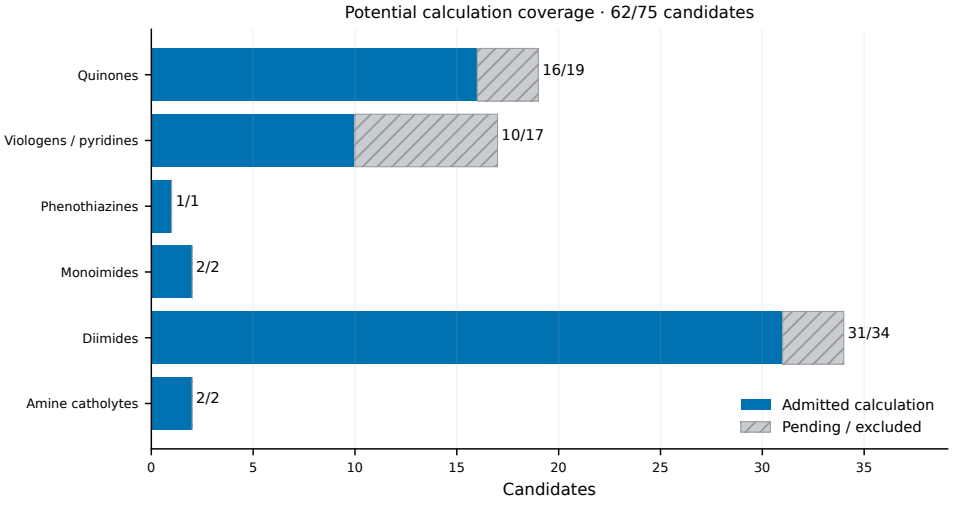
Defined hypotheses; source performance does not transfer

The study uses aqueous pH14/AgAgCl conditions, separate from MeCN/Fc. Isolated acid/phenol graphs and source chloride salts are recorded separately. Monoattachment, speciation and exact-graft synthesis still require review.

Nominal mass/capacity and SA only; no registration or quantum calculations. The largest graph has 70 atoms and needs a new UMA size/memory pilot.

Source: Rubio-Presa et al., 2025, DOI 10.1021/acsaem.5c02255. Preparation snapshot: 18 October 2026; two-electron ladders remain model hypotheses.

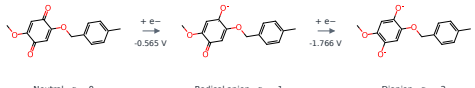
Family coverage across the complete candidate set



Formal redox-state drawings: computed capped models

Computed capped-model charge/spin ladders

1 Methoxy BQ

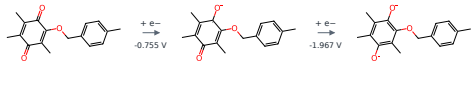


Neutral · q = 0
M = 1 (selected model)

Radical anion · q = -1
M = 2 (selected model)

Dianion · q = -2
M = 1 (selected model)

2 Trimethyl BQ

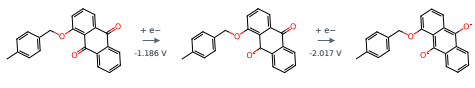


Neutral · q = 0
M = 1 (selected model)

Radical anion · q = -1
M = 2 (selected model)

Dianion · q = -2
M = 1 (selected model)

3 1-O AQ



Neutral · q = 0
M = 1 (selected model)

Radical anion · q = -1
M = 2 (selected model)

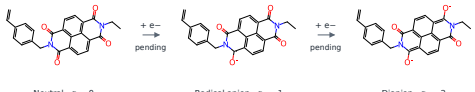
Dianion · q = -2
M = 1 (selected model)

Illustrative resonance forms; charge/spin may be delocalized. No proton transfer is implied.
Potentials: calculated MeCN vs Fc/Fc+. Arrows show electron count, not demonstrated reversibility.

Earlier exploration: vinyl-monomer redox drawings

Earlier vinyl exploration · proposed ladders

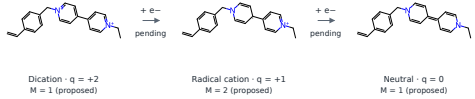
P1 NDI vinyl monomer



P2 PMDI vinyl monomer



P3 Viologen vinyl monomer



Illustrative resonance forms; charge/spin may be delocalized. No proton transfer is implied.
Earlier route assumption. Vinyl-state potentials remain pending; M is proposed.

Earlier route assumption: vinyl-monomer comparison

Core	Capped model	Vinyl monomer
	Nominal capacity / mAh g^{-1}	
NDI / ethyl	134.5	130.6
PMDI / ethyl	153.9	148.7
Viologen / ethyl	184.6	177.2

The larger open question is electronic

How much does the vinyl-bearing structure shift the redox potentials?

Same declared two-electron count; one added carbon. Stoichiometric capacities exclude counterions and polymer. Vinyl potentials remain pending.

Earlier exploratory results: historical context

Superseded reference and thermal treatment

The following figures and benchmark summaries retain the earlier calculation snapshot. They are not the updated candidate results or the current validation verdict.

Current results use repaired Fc geometries, uniform solution DFT thermochemistry, addressed caches and field-specific validation gates.

Historical redox benchmark (OROP)

Production-protocol OROP benchmark ($n = 169$):

	ours	OROP impl.-DFT
MAE (V)	0.52	0.28
signed bias	$+0.49$	—
Spearman (all)	0.92	—

At OROP's own level our organic potentials match theirs ($\sim$0.07 V); the gap sits in the Fc reference (4.19 vs 4.66 V). Sourced MeCN set: E1 bias $+0.26$ V, E2 -0.07 V. **Not calibrated away**.

Ranking must be judged *within* charge class

class	n	MAE	Spearman
$+1/0$ couples	118	0.48	**0.79**
$0/-1$ couples	49	0.60	**0.78**

Global correlation is inflated because charge classes separate. Ranking is assessed within each class; the integrity gate removes category errors.

The anion "failure" was category errors, not bad physics

Some apparent anion failures are category errors: molecules with **no intact, bound reduced state**:

molecule	exp	$\mathrm{RMSD}_{\mathrm{neu}\to\mathrm{an}}$
CCl_4	-2.57	**3.4 Å**
CH_2Br_2	-3.12	1.3 Å
(unbound anions)	—	$\mathrm{EA}_{\mathrm{gas}} < 0$

Dissociative electron attachment ($CCl_4 + e^- \to CCl_3^\bullet + Cl^-$) — scored against irreversible *peak* potentials.

Integrity screen

Bound anion (solvated EA > 0) *and* intact (**bond-graph unchanged**, not raw RMSD — which is confounded by tether flexibility).

- filtering → anion Spearman **0.49→0.95**
- a required necessary-condition gate and stability signal
- does **not** test proton transfer, dimerization, or long-term cycling

Stability = decomposition free energies

Disproportionation $2R \rightarrow R^+ + R^-$ (atom- & electron-conserving; ranks well, ρ 0.80, but **over-estimated by ~23 kJ/mol**):

intermediate	ΔG_{disp} (kJ/mol)
TEMPO$^\bullet$	$+228$
AQ semiquinone	$+82$
viologen MV$^{+\bullet}$	$+55$

TEMPO — the textbook persistent radical (our validation anchor) — ranks most stable, as it must. Exact from existing energies; no new calculation.

Dimerization $2\,\text{MV}^{+\bullet} \rightarrow [\text{MV}_2]^{2+}$ (the real viologen failure mode):

$\Delta G_{\text{dim}} = +0.49$ eV (flagged)

Config sampling: minimum well-defined ($\sigma_{\text{config}} \approx 2$ meV).
The uncertainty is the $+2$ **pimer desolvation** — the continuum's weak spot.

$\Rightarrow$ use dimerization as a **relative** ranking; lean on disproportionation for **ranking** (absolute values over-estimated, FINDINGS #24).

Kinetics: inner- and outer-sphere reorganization

Inner-sphere λ_i — solute geometry relaxation, **4-point (Nelsen)**:

$$\lambda_i = [E_O(q_R) - E_O(q_O)] + [E_R(q_O) - E_R(q_R)]$$

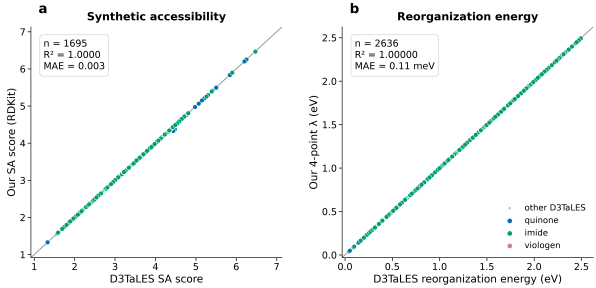
gas single points; same four-point formula as D3TaLES. **Validated**.

Outer-sphere λ_o — solvent reorganization (Marcus/Born, **Sharma Eq. 13**):

$$\lambda_o = \frac{z^2 e^2}{8\pi\varepsilon_0 r_{\mathrm{hyd}}} \left(\frac{1}{\varepsilon_{\mathrm{op}}} - \frac{1}{\varepsilon_r} \right)$$

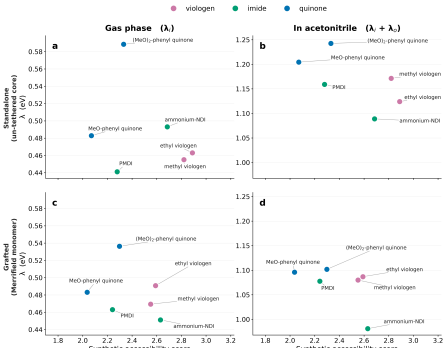
MeCN (ε_r=37.5, $\varepsilon_{\mathrm{op}}$=1.81); the production annotation uses a molecular-cavity nonequilibrium PCM, with Born as a check.

Report conventions explicitly:
$\lambda_{\mathrm{het}} = (\lambda_O + \lambda_R)/2 + \lambda_{o,1}$;
$\lambda_{\mathrm{SE}}(d) = \lambda_O + \lambda_R + 2\lambda_{o,1}(1 - a/d)$.



At D3TaLES's own level, inner-sphere λ agrees to **median** $|\Delta| = 0.09$ **eV** ($n = 322$).

Candidates: SA vs. reorganization energy

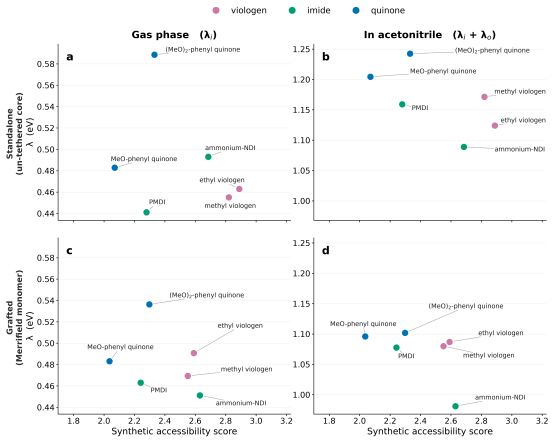


All six are *grafted* model-compounds (redox core + 4-methylbenzyl monomer proxy); short class names label them.

- one point per molecule (mean over its charge states); the six cluster tightly ($\lambda_i \approx$0.45–0.54 eV — similar across families).
- **ammonium-NDI**: lowest $\lambda \Rightarrow$ fastest ET; **quinones** highest.
- gas $\rightarrow$ MeCN: $+\lambda_o$ lifts each point $\sim$0.5–0.6 eV (note per-panel y-zoom).
- λ_i validated (D3TaLES R^2=1.0); λ_o Marcus/Born estimate (r_{hyd} caveat) — read *relative* trends.

Historical snapshot: superseded reference/thermals; current validation does not transfer

Functionalization × solvation: the 2×2



Rows = **before** (standalone core) vs **after** functionalization (grafted: redox core + 4-methylbenzyl monomer handle).
Cols = gas (λ_i) vs MeCN ($\lambda_i+\lambda_o$).

Functionalization barely moves λ_i — $\pm$15–50 meV, mixed sign:

- viologens: $+15$ / $+29$ meV
- PMDI $+22$; **NDI** $\mathbf{-44}$; quinones 0 / $\mathbf{-52}$

Solvation adds a near-uniform **+0.5–0.6 eV** (λ_o).

$\Rightarrow$ the **redox core sets** λ; the monomer handle is a small perturbation, so the cheap standalone λ is a good predictor of the grafted value.

What we learned (the durable lessons)

What works

- ranking (within charge class)
- xtb RRHO thermal (open-shell safe)
- disproportionation stability
- reorganization energy λ
- intact/bound state as a necessary-condition gate
- capacity (n, MW): exact

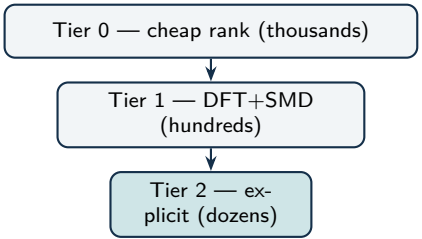
Negative results (saved effort)

- continuum ion-pairing: **refuted** (spacing off by $\sim$11 V)
- explicit microsolvation (4 MeCN): **no gain** for viologen
- absolute redox offset: **unresolved** (OROP vs in-house anchors)
- dimerization absolute: **unreliable** ($+2$ desolvation)
- solubility / crossover: proxies only

Corroborated by the metabolic-ΔG pipeline: 0.14 eV via isodesmic cancellation $\Rightarrow$ **cancellation, not more physics, is the lever**.

Putting it together: a tiered, multi-objective search

Tiered funnel (explicit solvation is a finisher, never for thousands):



Pareto scorecard — built correctly

- objectives = **trustworthy** axes only (capacity, λ, disprop. stability, ranked potential)
- proxies (solubility, SA, crossover) = **filters**, not axes
- **integrity** = hard necessary-condition pre-filter
- missing applicable data = **INCOMPLETE**, never a favorable value
- split **anolyte / catholyte** (potential isn't "more is better")
- σ-**aware** domination (don't rank within error)
- read the front against the *raw* axes — no scalarized score (arbitrary weights, σ-blind, pool-relative)

Applying it: candidate scorecard → shortlist

σ-aware Pareto on the current rankable grafted set; rows missing an applicable objective are **INCOMPLETE** and cannot enter or dominate the primary front:

id	E_{an}	Cap	λ	ΔG_d	Par.
aq_benzyloxy*	−1.46	163	0.39	61	**Y**
pmdi	−1.46	160	0.47	64	—
mophquinone*	−1.02	176	0.52	92	**Y**
dmophquinone	−1.00	147	0.57	89	—
viologen†	−0.94	194	0.47	51	TBA
aq_benzylamino‡	−1.57	164	—	76	partial

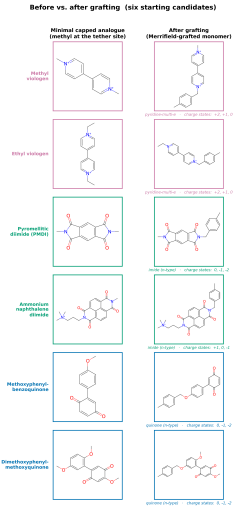
Ordered by primary axis (E_{an}). E_{an} [V vs Fc], Cap [mAh/g], λ [eV], ΔG_d (disprop.) [kJ/mol].

* = primary-front under both salts; † joins under $TBAPF_6$; ‡ is a partial front only (no QC-clean λ). The front and raw axes are reported without an arbitrary scalar score.

Completeness matters: a missing or QC-flagged λ cannot make a candidate look artificially fast. The row remains visible as a partial front, with its missing objective.

⇒ across this tight set voltage is *non-discriminating*; **capacity + disproportionation stability + λ** drive the ranking.

Current focus: six multi-electron starting candidates



Two **viologens**, two **imides**, and two **quinones**: capped core (left) versus Merrifield-grafted proxy (right).

Historical snapshot: superseded reference/thermals; current validation does not transfer

Next steps

Done: protocol-addressed energies; unified σ-tagged scorecard; integrity and completeness contracts; σ-aware Pareto engine.

1. **Resolve the E° offset** — cross-run in-house and OROP molecules; audit OROP's thermodynamic cycle and reference convention before calibration.
2. **Broaden benchmarks** — second reductions and at least ten two-wave molecules.
3. **Complete kinetics** — conformer-matched core λ for tether jumps and solvated λ where a gas dianion is unbound.
4. **Polymer effects** — backbone/tether constraints & inter-monomer dimerization (where the *relative* dimerization ranking becomes key).
5. **Absolute accuracy** (if needed) — same-charge referencing to remove the per-class redox bias.

Historical summary

We can **rank** redox candidates and screen **stability**, **kinetics**, and **state integrity** on a physics-based footing — with each axis **honestly labelled** by how far we can trust it.

The σ-aware scorecard & Pareto engine are built and applied. Next: resolve absolute E° and close the incomplete kinetics rows before scaling.

Historical snapshot; current candidate and property validation remains separate.